Computational Statistical Physics
Exercise 7

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1 Description

This report aims to illustrate and discuss results from diatomic molecular dynamics simulations as well as a canonical ensemble regulated by a Nosé-Hoover thermostat. Code used to generate results discussed in this report can be accessed via the following link.

1.1 Intra-molecular forces

For particles not composing the same molecule, dynamics are described by the Verlet scheme accordingly

$$\mathbf{x}_i(t+\Delta t) = 2\mathbf{x}_i(t) - \mathbf{x}_i(t-\Delta t) + \Delta t^2 \frac{\mathbf{f}_i}{m} \tag{1}$$

where $\mathbf{f}$ may describe any kind of n-body force acting on each particle. In this case, $\mathbf{f}$ results from the Lennard-Jones potential, taking into account the attraction/repulsion between particles. For particle P_i, it is given by the following equations:

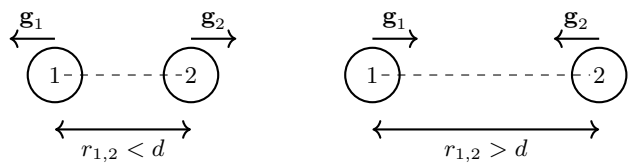
$$\begin{cases} \mathbf{f}_i = -\sum_j \nabla V(r_{i,j}) \\ V(r) = 4\sigma\left[\left(\frac{1}{r}\right)^{12} - \left(\frac{1}{r}\right)^{6}\right] \end{cases}$$

Periodic boundary conditions are used at the boundaries of the volume containing the particles and a cut-off radius r_c is defined beyond which no Lennard-Jones interaction is calculated.

1.2 Molecular Constraints

We consider the constraint imposed by fixing two particles P_1 and P_2 to remain at the given distance $r_{1,2} = d$. Essentially, considering the particles as part of a di-atomic molecule, corrective inter-molecular forces $\mathbf{g}_1$, $\mathbf{g}_2$ describes the forces aiming to maintain the magnitude of the particles relative position at d,

after some disruption event. This being the sole objective, the inter-molecular forces act only along the axis passing through the center of both particles, as illustrated below.



Based on the figure, it is edvident that the forces can be expressed as a multiple of the relative position vector, yielding the following set of equations.

$$\begin{cases} \mathbf{g}_1 = \lambda \mathbf{r}_{1,2} \\ \mathbf{g}_2 = \lambda \mathbf{r}_{2,1} = -\lambda \mathbf{r}_{1,2} \end{cases}$$

For every iteration, update steps are performed taking into account external/intra-molecular forces. Equation (1) is re-expressed, this time with a tilde denoting that another update step is yet to be performed, taking into account the molecular constraint.

$$\tilde{\mathbf{x}}_i(t + \Delta t) = 2\mathbf{x}_i(t) - \mathbf{x}_i(t - \Delta t) + \Delta t^2 \frac{\mathbf{f}_i}{m} \tag{2}$$

Afterwards, the inter-molecular force is applied to correct for the possible miss-match in the particle displacement relative to it's pair particle, especially prone to happen for particles rotating relative to one-another.

$$\mathbf{x}_i(t + \Delta t) = \tilde{\mathbf{x}}_i(t + \Delta t) + \Delta t^2 \frac{\mathbf{g}_i}{m} \tag{3}$$

Finally, for the particles $P_1,\ P_2$ previously considered, the Lagrange factor λ can be derived via the following constraint equation

$$\begin{aligned} d &= |\mathbf{x}_1(t + \Delta t) - \mathbf{x}_2(t + \Delta t)| \\ &= |\tilde{\mathbf{x}}_1(t + \Delta t) + \Delta t^2 \frac{\mathbf{g}_1}{m} - \tilde{\mathbf{x}}_2(t + \Delta t) - \Delta t^2 \frac{\mathbf{g}_2}{m}| \\ &= |\tilde{\mathbf{r}}_{1,2}(t + \Delta t) + \Delta t^2 \frac{1}{m}\big(\lambda \mathbf{r}_{1,2} - \lambda \mathbf{r}_{2,1}\big)| \\ &= |\tilde{\mathbf{r}}_{1,2}(t + \Delta t) + \Delta t^2 \frac{2\lambda}{m} \mathbf{r}_{1,2}(t)| \end{aligned}$$

Abbreviating the time-dependency and squaring both sides:

$$\left(\tilde{\mathbf{r}}_{1,2} + \Delta t^2 \frac{2\lambda}{m} \mathbf{r}_{1,2}\right)^2 = d^2$$

Expanding terms:

$$|\tilde{\mathbf{r}}_{1,2}|^2 + 2\lambda \cdot \frac{2\Delta t^2}{m}\tilde{\mathbf{r}}_{1,2} \cdot \mathbf{r}_{1,2} + \lambda^2 \cdot \left(\frac{2\Delta t^2}{m}\right)^2 |\mathbf{r}_{1,2}|^2 = d^2$$

This is a quadratic equation in λ:

$$\lambda^2 \cdot \left(\frac{2\Delta t^2}{m}\right)^2 |\mathbf{r}_{1,2}|^2 + 2\lambda \cdot \frac{2\Delta t^2}{m}\tilde{\mathbf{r}}_{1,2} \cdot \mathbf{r}_{1,2} + |\tilde{\mathbf{r}}_{1,2}|^2 - d^2 = 0$$

Solve using the quadratic formula:

$$\lambda = \frac{\sqrt{(\tilde{\mathbf{r}}_{1,2} \cdot \mathbf{r}_{1,2})^2 - |\mathbf{r}_{1,2}|^2 \left(|\tilde{\mathbf{r}}_{1,2}|^2 - d^2\right)} - \tilde{\mathbf{r}}_{1,2} \cdot \mathbf{r}_{1,2}}{\frac{2\Delta t^2}{m} |\mathbf{r}_{1,2}|^2} \tag{4}$$

Which yields the factor scaling the relative position in the inter-molecular update equation (2).

1.3 Nosé-Hoover Thermostat

Intending to simulate a canonical ensemble with fixed temperature T, a Nosé-Hoover thermostat is used to "speed up" or "slow down" individual particles. The physical reasoning behind the thermostat involves introducing a variable s, scaling the systems time evolution, as if it was connected to a heat bath or reservoir. Via this procedure, each particle adheres to maintaining an average kinetic energy of $3/2\ k_B T$, consistent with the number of degrees of freedom for single particles in 3D. The extended Hamiltonian for the Nosé thermostat is:

$$H = \sum_i \frac{\mathbf{p}_i^2}{2ms^2} + \sum_{i,j} V(\mathbf{r}_i - \mathbf{r}_j) + \frac{p_s^2}{2Q} + gk_B T \ln s \tag{5}$$

This is reformulated into an update equation by introducing a friction variable $\zeta = \dot{s}/s$:

$$\zeta = \frac{1}{Q}\left(\sum_i \frac{\mathbf{p}_i^2}{m} - gk_B T\right), \tag{6}$$

With $g = 3N + 1$ describing the number of degrees of freedom, and Q the coupling strength between the system and the reservoir. Finally, ζ is incorporated into the system dynamics by acting as a force on each and every particle in the following way:

$$\dot{\mathbf{p}}_i = \mathbf{f}_i \ + \ \zeta \mathbf{p}_i \tag{7}$$

This is here implemented numerically in a different manner than the update equations described thus far. Details can be found in a document from The University of Edinburgh[1]

[1] `https://www2.ph.ed.ac.uk/~dmarendu/MVP/MVP03.pdf`

2 Molecular Dynamics Simulation

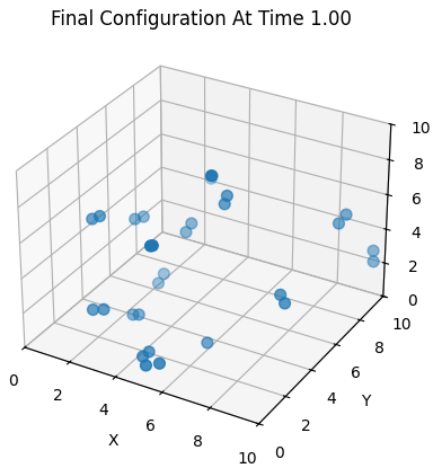


Figure 1: Snapshot of Molecular Dynamics simulation.

Particles are initialized in pairs with distance d apart, each pair representing a molecule. 30 particles, i.e. 15 molecules are simulated over a total time $t = 1$, with time-steps $\Delta t \in \{10^{-3}, 10^{-4}\}$. Over the course of the simulation, energy conservation is considered to check the validity of the implementation.

2.1 Total Energy Conservation

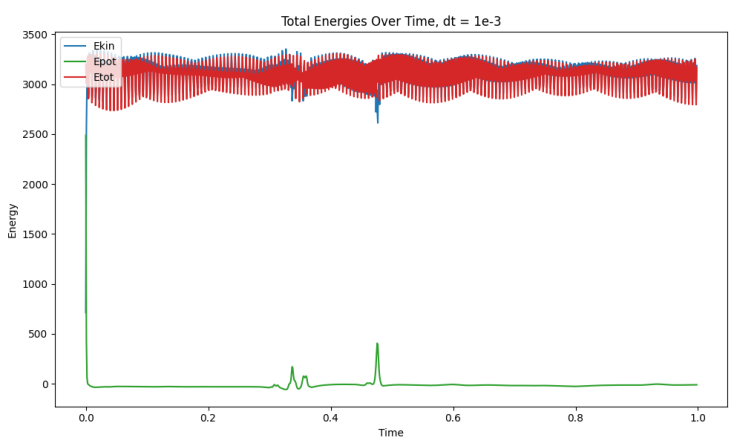


Figure 2: Molecular Dynamics simulation, simulated for time-steps $\Delta t = 10^{-3}$.

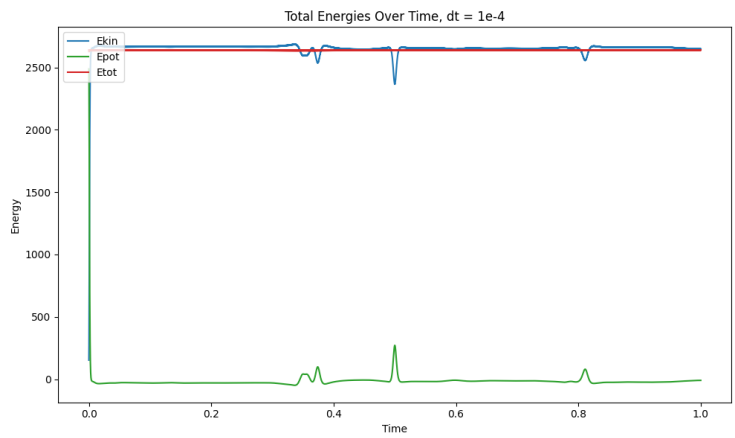


Figure 3: Molecular Dynamics simulation, simulated for time-steps $\Delta t = 10^{-4}$.

For the larger time-step, the system shows rapidly oscillatory behavior in terms of the total kinetic energy E_{kin}.

2.2 Stability Analysis of MD Simulation

To analyze the oscillatory behavior further, the total bond energy E_{bond} for the system is considered, denoting the sum of each molecular bond energy, in term given as:

$$\epsilon_{bond} = \frac{1}{2}\lambda(r_{i,j} - d)^2 \tag{8}$$

for a molecule M consisting of particles $P_i,\ P_j$. Also, we now consider the Fourier domain, as the oscillatory behavior is best represented as a frequency.

As expected, for the smaller time-step, the total kinetic energy shows strong peaks at multiples of the total bond energy frequencies. For the larger time-step, the bond energy peaks persists, but this time with less corelation to the kinetic energy peaks. This shows that a sufficiently small time-step must be picked to ensure physically reliable behavior for molecules, perhaps smaller than for the corresponding configuration for single particles.

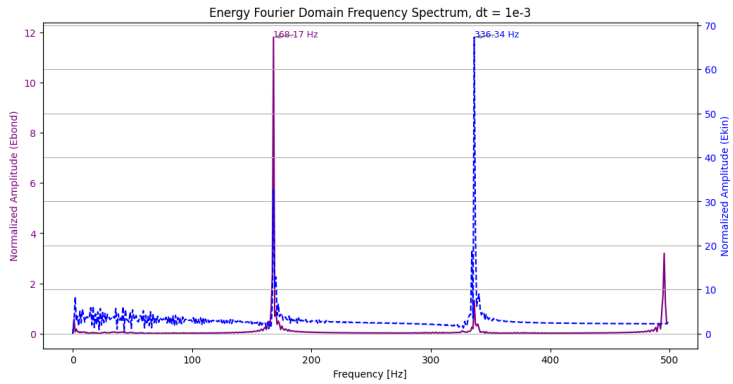


Figure 4: Fourier transforms of total kinetic energy and total bond energy for time-step $\Delta t = 10^{-3}$.

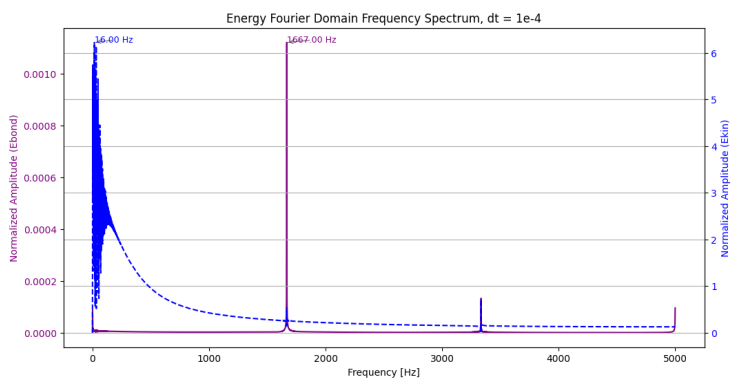


Figure 5: Fourier transforms of total kinetic energy and total bond energy for time-step $\Delta t = 10^{-4}$.

3 Canonical Ensemble Simulation

In the following simulations, canonical ensembles at temperature $T = 10$ are simulated. The Boltzmann constant k_B is always assumed to equal 1.

3.1 Instantaneous Temperature and Total Energy

Simulating the canonical ensemble, thermostats with three different coupling strengths are simulated over a time span $t = 10$ with the same particle number and temperature, namely $Q \in \{10, 10^3, 10^6\}$. First is considered the instantaneous temperature, resulting from the sum over the particle kinetic energies:

$$T_{inst} = \frac{2}{k_B \cdot 3(N-1)} \sum_i \epsilon_{k,i} \tag{9}$$

where from now one, we rely on the short-hand notation ϵ_k to denote the particle kinetic energies.

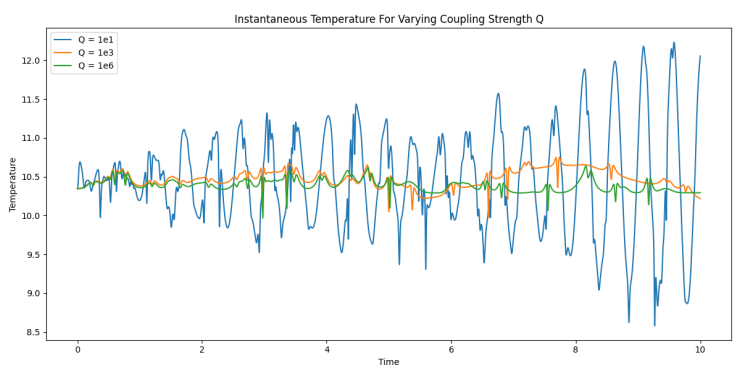


Figure 6: Instantaneous temperature as measured from the canonical ensemble.

For the lowest value on Q, the system shows strong oscillations. This is not a lack of numerical stability of the system, but instead a direct feature of the Nosé-Hoover thermostat. Readers familiar with control theory would quickly relate the thermostat to a so called "PID" regulator, although with only the integrator coefficient being non-zero. As such, the thermostat will consistently overshoot when trying to regulate back to the desired temperature. This oscillatory behavior is minimized for larger Q. With Q being sufficiently large, as in the case $Q = 10^6$, it is evident from the total energy plot that the system resembles more of a micro-canonical ensemble. This is characterized by the total energy being near constant.

3.2 Particle Distribution of Eqilibrated Ensemble

This section aims to illustrate the kinetic energy distribution after a certain number of time-steps, to compare with theoretical derivations on canonical en-

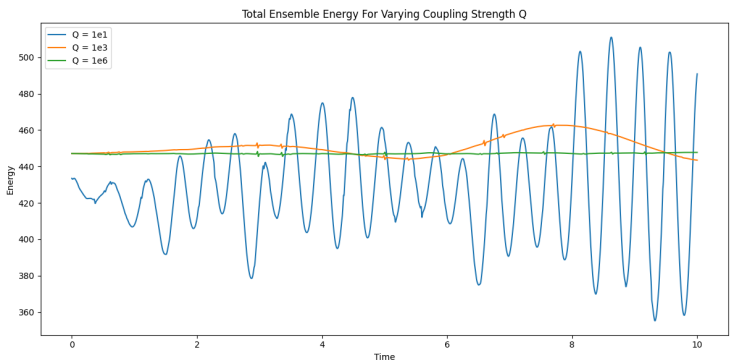


Figure 7: Total kinetic energy as measured from the canonical ensemble.

sembles. Therefore, it is first important to recognize the particle kinetic energy density, given as:

$$f(\epsilon_{kin}) = \frac{2\beta}{\sqrt{\pi}}\sqrt{\beta\epsilon_k}e^{-\beta\epsilon_k} \tag{10}$$

in this case normalized to 1 when integrated over the possible kinetic energies $\epsilon_k \in [0, \infty]$. Also, as most readers probably realize $\beta = 1/(k_B T)$.

The theoretical density will in turn be compared to a parametrized curve of the form:

$$g(A, B, x) = A\sqrt{Bx}e^{-Bx} \tag{11}$$

which will be fitted against the generated kinetic energy distribution to yield the coefficients A, B.

The particle data is generated by increasing the particle number to $N = 500$ and running until $t = 20$. Important to take into consideration here, is the particle kinetic energies being initialized uniformly, but of course the initial velocity vector pointing in arbitrary directions. The kinetic energy distribution is considered for a few different times across the simulation.

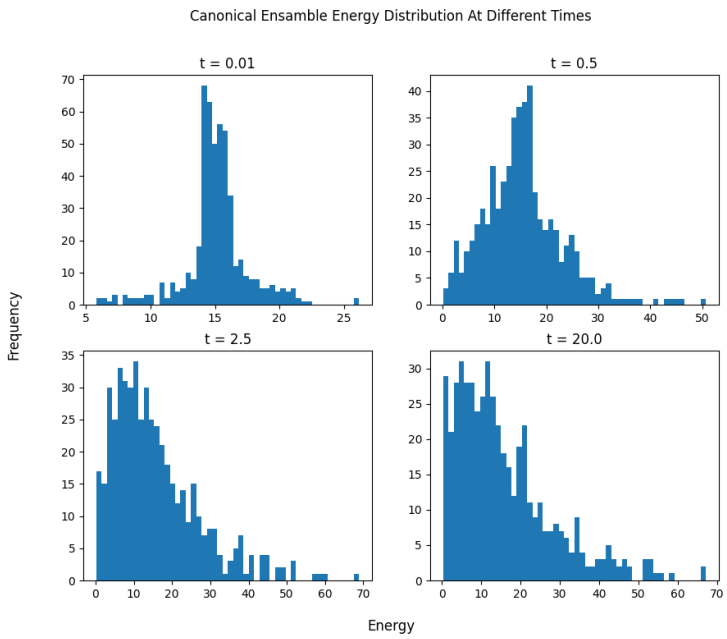


Figure 8: Kinetic energy distribution of canonical ensemble at times $t = 0.01, 0.5, 2.5$ & 20.

Bringing our attention to the final time $t = 20$, it is apparent from the shape that this could very likely be an accurate representation of the equilibrated particle distribution. Fitting the parametrized curve and comparing with the theoretical density yields the following:

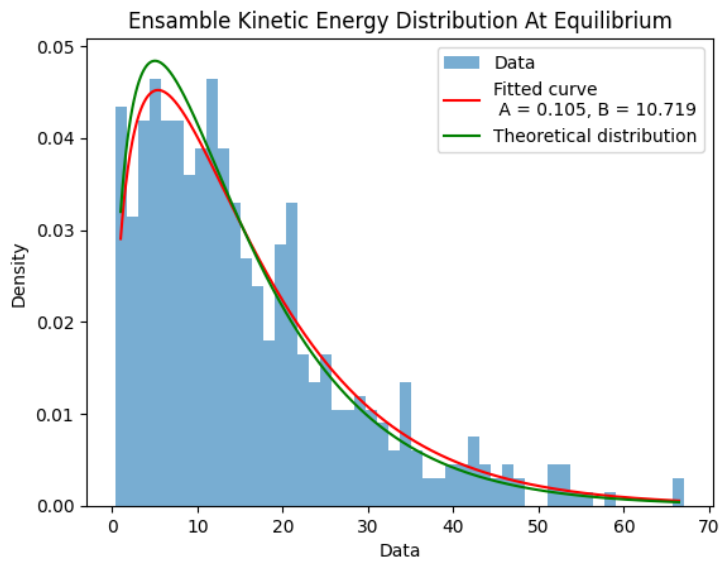


Figure 9: Data analysis and theoretical comparison for the distribution at time $t = 20$.

The resulting coefficients $A = 0.105$ and $B = 10.719$ do not deviate a substantial amount from the corresponding "true" coefficients $2\beta/\sqrt{\pi} = 0.113$ and $\beta = 10$. This result verifies that the Nosé-Hoover thermostat can indeed be used to accurately simulate canonical ensembles.